

RLAIF vs. RLHF: Scaling Reinforcement Learning from Human Feedback with AI Feedback

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Introduction

Reinforcement Learning from Human Feedback (RLHF)

Reinforcement Learning from Human Feedback (RLHF) is an effective technique for aligning language models to human preferences. It is cited as one of the key drivers of success in modern conversational language models, such as ChatGPT (Liu et al., 2023) and Bard (Manyika, 2023)

A key advantage of training language models with reinforcement learning (RL) is that it enables optimization on complex, sequence-level objectives that are not easily differentiable and therefore ill-suited for traditional supervised fine-tuning (SFT)

One obstacle for employing RLHF at scale is its dependence on high-quality human preference labels.

Introduction

Reinforcement Learning from AI Feedback (RLAIF)

Bai et al.(2022b) was the first effort to explore Reinforcement Learning from AI Feedback (RLAIF), where RL was conducted using a reward model trained on a hybrid of human and AI preferences.

In conjunction with their “Constitutional AI” self-revision technique, their final policy outperformed supervised fine-tuning for training a conversational assistant.

Introduction

RLHF vs. RLAIF

In this work, we compare the effectiveness of RLAIF and RLHF (see Figure 2) on three tasks:

- Summarization
- Helpful dialogue generation
- Harmless dialogue generation

The win rates for RLAIF and RLHF are not statistically significantly different.

Furthermore, in a head-to-head comparison of RLAIF against RLHF, both policies are equally preferred.

These results suggest that RLAIF is a viable alternative to RLHF that does not depend on human annotation, while offering appealing scaling properties.

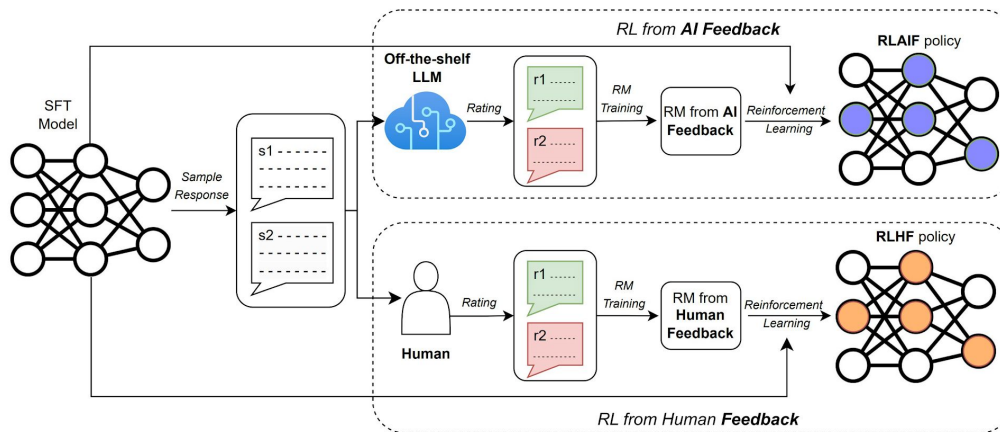
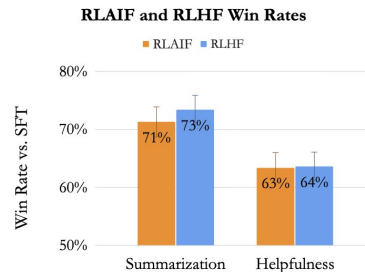
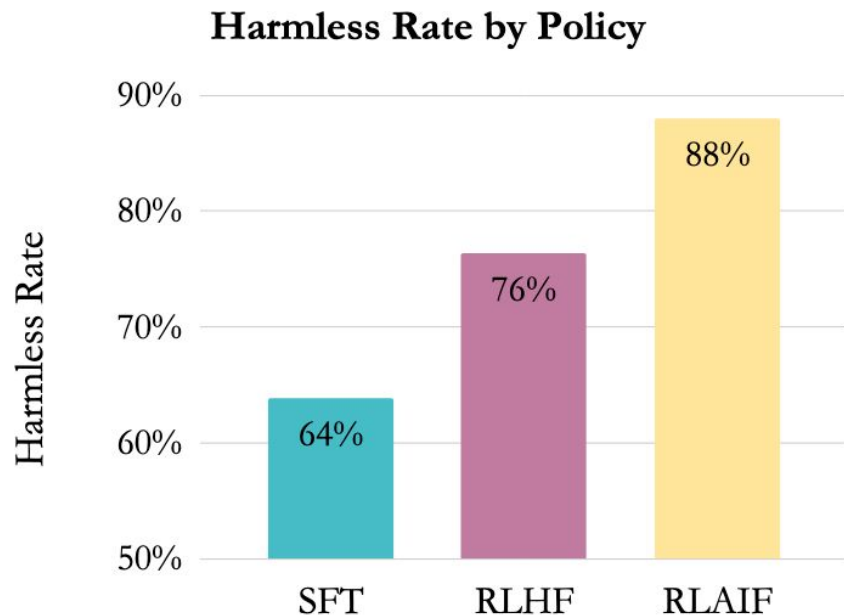


Figure 2: A diagram depicting RLAIF (top) vs. RLHF (bottom)

Introduction

SFT vs. RLAIIF

First, in a step towards LLM self-improvement, we demonstrate that RLAIIF significantly improves upon the SFT baseline even when the AI labeler is the same size as the policy model.



Introduction

d-RLAIF

Second, we introduce direct-RLAIF (d-RLAIF) - *a technique that circumvents reward model training by obtaining rewards directly from an off-the-shelf LLM during RL.*

- d-RLAIF matches or outperforms canonical RLAIF
- for the task of helpful dialogue generation, the initial policy and the LLM providing rewards are the same model checkpoint, demonstrating an instance of strict LLM self-improvement.

Introduction

Techniques to maximize the alignment of AI-generated preferences to human preferences

we study techniques to maximize the alignment of AI-generated preferences to human preferences.

We find that soliciting chain-of-thought reasoning (Wei et al., 2022) consistently improves alignment, while using a detailed preamble and few-shot prompting (Brown et al., 2020) are only beneficial for certain tasks.

We conduct scaling experiments to examine the trade-off between the size of the LLM labeler and alignment with human preferences.

Methodology

Preference Labeling with LLMs

1. Preamble

Introduction and instructions describing the task at hand

2. Few-shot exemplars (optional)

An example input context, a pair of responses, a chain-of-thought rationale (optional), and a preference label

3. Sample to annotate

An input context and a pair of responses to be labeled

4. Ending

The ending text to prompt the LLM (e.g. “Preferred Response=”)

Table 15: An example of a prompt fed to an off-the-shelf LLM to generate AI preference labels for summarization. {text}, {summary1}, and {summary2} are populated with unlabeled examples, and a preference distribution is obtained by computing the softmax of the log-probabilities of generating the tokens “1” vs. “2”.

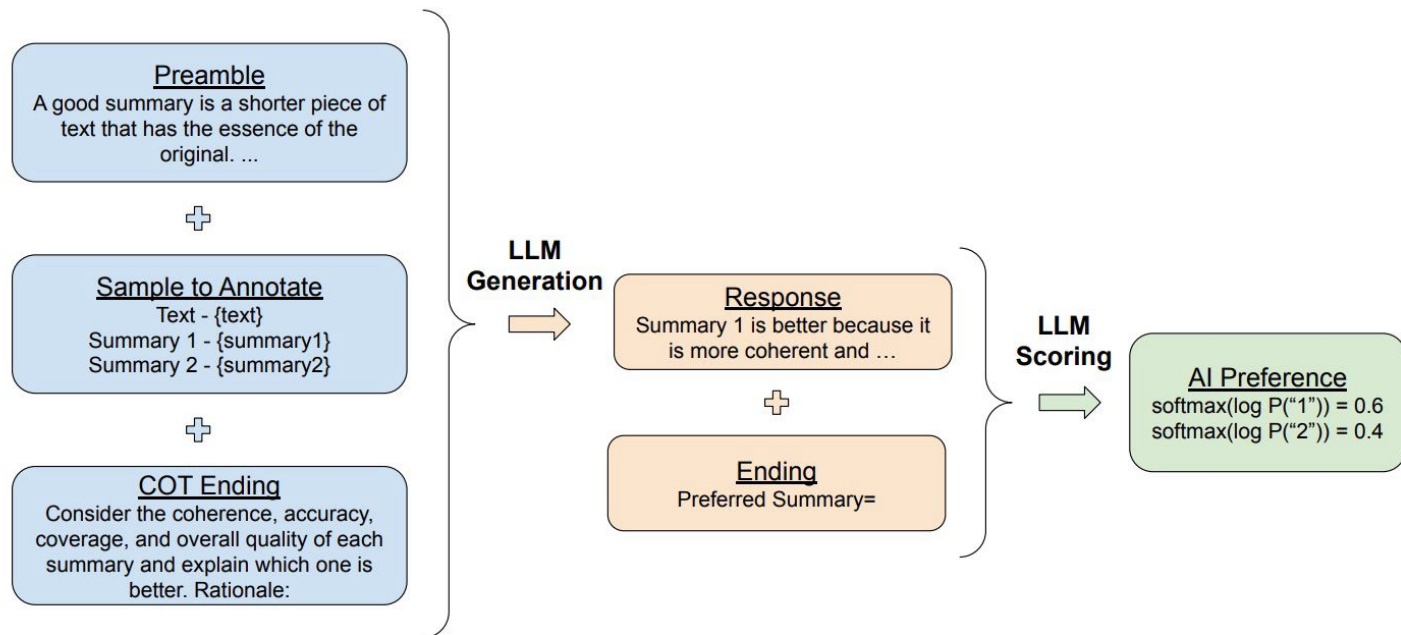
Preamble	A good summary is a shorter piece of text that has the essence of the original. ... Given a piece of text and two of its possible summaries, output 1 or 2 to indicate which summary best adheres to coherence, accuracy, coverage, and overall quality as defined above.
Exemplar	»»»» Example »»»» Text - We were best friends over 4 years ... Summary 1 - Broke up with best friend, should I wish her a happy birthday... And what do you think of no contact? Summary 2 - should I wish my ex happy birthday, I broke no contact, I'm trying to be more patient, I'm too needy, and I don't want her to think I'll keep being that guy. Preferred Summary=1 »»»» Follow the instructions and the example(s) above »»»»
Sample to Annotate	Text - {text} Summary 1 - {summary1} Summary 2 - {summary2}
Ending	Preferred Summary=

we extract the log-probabilities of generating the tokens “1” and “2” and compute the softmax to obtain a preference distribution.

Methodology

Preference Labeling with LLMs

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Methodology

Preference Labeling with LLMs - **Addressing Position Bias**

We find evidence of position bias, which is especially prevalent in smaller LLM labelers

Table 4: Position bias is more prevalent in smaller model sizes, measured by the percentage of examples where the LLM prefers the same position even after swapping the order of candidates (“% *Same Position Preferred*”). Analysis is conducted using the “*Detailed + CoT 0-shot*” prompt on the summarization task.

Model Size	% Same Position Preferred
PaLM 2 L	18%
PaLM 2 S	21%
PaLM 2 XS	56%

Two inferences are made for every pair of candidates, where the order in which candidates are presented to the LLM is reversed for the second inference.

The results from both inferences are then averaged to obtain the final preference distribution.

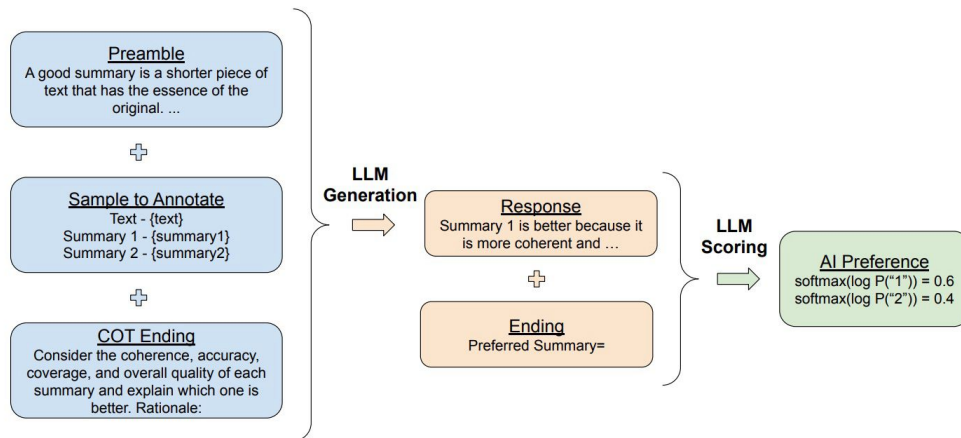
Methodology

Preference Labeling with LLMs - **ELICITING CHAIN-OF-THOUGHT REASONING**

We experiment with eliciting chain-of-thought (CoT) reasoning (Wei et al., 2022) from our AI labelers through a two-step inference procedure.

1. Replace the Ending of the standard prompt with a sentence asking for thoughts and explanation
2. Decode a response from the LLM.

Then, we concatenate the original prompt, the response, and the standard Ending string together, and follow the scoring procedure



Methodology

Reinforcement learning setups - **CANONICAL RLAI**

A.2. Reward Modeling

Given an input x , we sample a pair of responses $(y_1, y_2) \sim \pi$ from one or more models, where oftentimes π is the SFT model. The input and responses are sent to human annotators to rate which response is better according to some criteria. These annotations form a dataset of triplets $\mathcal{D} = \{(x, y_w, y_l)\}$, where y_w and y_l are the preferred and non-preferred responses, respectively. A reward model (RM) r_ϕ is trained by minimizing the following loss:

$$\mathcal{L}_r(\phi) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma(r_\phi(x, y_w) - r_\phi(x, y_l)) \right],$$

where σ is the sigmoid function.

A reward model (RM) is trained on the LLM-generated preference label

We train the RM with a cross-entropy loss on the softmax of the scores generated by the RM.

The softmax converts the RM scores into a probability distribution.

Training a RM on a dataset of AI labels can be viewed as a form of model distillation.

Methodology

Reinforcement learning setups - **DIRECT-RLAIF (D-RLAIF)**

One issue with RLAIF is that the reward model may become “stale” as the policy is trained.

As the policy is trained, the generated trajectories become increasingly out-of-distribution from the dataset the RM was trained on, leading to suboptimal performance (Bai et al., 2022a).

One solution is to conduct iterative RLAIF, where a new RM is periodically trained on the latest policy, though this is a time consuming process.

Methodology

Reinforcement learning setups - **DIRECT-RLAIF (D-RLAIF)**

direct-RLAIF (d-RLAIF) - a simple alternative to canonical RLAIF that directly uses LLM feedback as the reward signal in RL.

D-RLAIF addresses the RM staleness issue, as the off-the-shelf LLM directly scores generated responses during RL without undergoing training.

Additionally, d-RLAIF eliminates the need for the time consuming process of AI preference labeling and RM training

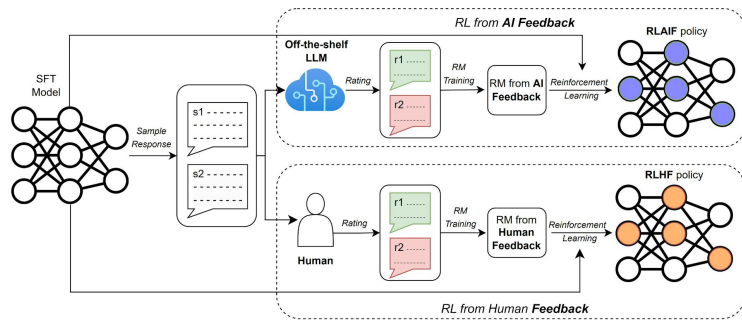
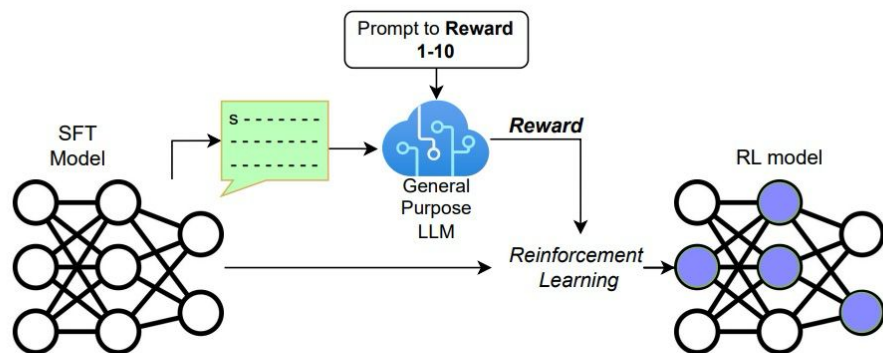


Figure 2: A diagram depicting RLAIF (top) vs. RLHF (bottom)



Methodology

Evaluation Metrics - **AI Labeler Alignment**

AI Labeler Alignment measures the accuracy of AI-labeled preferences with respect to human preferences

The alignment accuracy z_{acc} can be expressed as follows:

$$z_{acc} = \frac{1}{D} \sum_{i=1}^D \mathbb{1}[\arg \max_j P_{i,j}^{AI} = p_i^H],$$

where D is the size of the preference dataset, $P^{AI} \in \mathbb{R}^{D \times 2}$ is the matrix of soft AI preferences, and $p^H \in \mathbb{R}^D$ is the corresponding vector of human preferences, containing elements 0 or 1 to denote whether the first or second response is preferred, respectively.

Methodology

Evaluation Metrics - **Win Rate**

Win Rate evaluates the end-to-end quality of two policies by measuring how often one policy is preferred by human annotators over another.

Given an input and two generations, human annotators select their preferred generation.

The percentage of instances where policy A is preferred over policy B is referred to as the “win rate of A vs. B”. A 50% win rate indicates that A and B are equally preferred.

Methodology

Evaluation Metrics - **Harmless Rate**

Harmless Rate measures the percentage of responses that are considered harmless by human evaluators.

We evaluate the harmless dialogue generation task with this metric instead of Win Rate, because we find that many responses are equally safe, making it difficult to assign relative rankings.

Experimental Details

Datasets

- Reddit TL;DR (Stiennon et al., 2020)
- OpenAI's Human Preferences (Stiennon et al., 2020)
- Anthropic Helpful and Harmless Human Preferences (Bai et al., 2022a)

Experimental Details

LLM Labeling

- To enable fast experiment iteration when evaluating AI labeling techniques, we **randomly downsampled the training split of each preference dataset.**
- For summarization, **an additional filter was applied to only include examples where human annotators preferred one summary over the other with high confidence,**
- After downsampling and filtering, there remained 3-4k examples for each task
- AI labeler alignment was calculated on these downsampled datasets with PaLM 2 Large.

Experimental Details

Model Training

- All SFT models are initialized from PaLM 2 Extra-Small (XS)
- For summarization, the SFT model is produced by fine-tuning PaLM 2 XS on the Reddit TL;DR dataset.
- All RMs are trained from PaLM 2 XS checkpoints.
- RM is fine-tuned on the full training split of a preference dataset, where the label is the AI preference for AI feedback RMs and the original human preference label for human feedback RMs.
- we use REINFORCE with a baseline given that it is simpler yet still effective for the problem at hand.

Experimental Details

Human Evaluation

o measure win rates, evaluators were presented with an input context and multiple responses generated from different policies (e.g. RLAIIF, RLHF, and SFT). They were then asked to rank responses in order of quality without ties

Context

I'm guessing that most people today have some form of DVR, whether it be a Tivo or the one you rent from your local television service provider. I personally have a Ceton InfiniTV 4 card in my computer with 2 Switched Digital Video Tuning Adapters that provides for 4 HD streams at a time. That along with a 2 TB Hard Drive makes for a pretty decent queue of recorded television shows for my fiancé and I to watch when we feel like it. Today I sat down on the couch and after flipping through the recorded TV queue decided I didn't really want to watch any of those things, so I hit the "Guide" button on my remote to see what was on. It dawned on me that this was the first time in at least 2 months that I had watched live television, when I tried to skip forward past a commercial and realized that I couldn't lol. So yeah, how about you?

[Clear highlighted text](#) [Clear response ranking](#)

Model outputs

I have a DVR and I haven't watched live television in at least 2 months. When was the last time you watched live television?

#4 best response

I have a DVR and I realized today that I haven't watched live television in at least 2 months. When was the last time you watched live TV?

#2 best response

Realized that I'd been watching nothing but DVR'd TV for at least two months. When is the last time you actually watched live television?

✓ Best response

I haven't watched live television in at least 2 months, and I'm wondering how long it's been since you've done the same.

#3 best response

Figure 5: A screenshot of the user interface presented to human evaluators, ultimately used to calculate win rates. Raters are shown a context and asked to rank the quality of candidate responses.

Results

RLAIF vs. RLHF

- Specifically, RLAIF and RLHF are preferred by human evaluators over the baseline SFT policy 71% and 73% of the time for summarization⁶ and 63% and 64% for helpful dialogue generation, respectively.
- The difference in win rates between RLAIF vs. SFT and RLHF vs. SFT are not statistically significant.

Table 1: **Left side:** Win rates for pairs of policies on the summarization and the helpful dialogue tasks. **Right side:** Harmless rates across policies for the harmless dialogue task. All numbers are based on human evaluation.

Win Rate			Harmless Rate	
Comparison	Summa- rization	Helpful dialogue	Model	Harmless dialogue
RLAIF vs SFT	71%	63%	SFT	64%
RLHF vs SFT	73%	64%	RLHF	76%
RLAIF vs RLHF	50%	52%	RLAIF	88%
Same-size RLAIF vs SFT	68%	—		
d-RLAIF vs SFT	74%	66%		
d-RLAIF vs Same-size RLAIF	60%	—		

Results

Towards Self-Improvement

- The LLM used to label preferences (PaLM 2 L) is much larger than the policy being trained (PaLM 2 XS). Going one step further, we explore whether RLAIIF can yield improvements when the AI labeler is the same size. RLAIIF achieves a statistically significant improvement over RLHF and SFT, according to two-sided paired t-tests. as the policy. We refer to this setup as “same-size RLAIIF”.

Table 1: **Left side:** Win rates for pairs of policies on the summarization and the helpful dialogue tasks. **Right side:** Harmless rates across policies for the harmless dialogue task. All numbers are based on human evaluation.

Win Rate			Harmless Rate	
Comparison	Summa- rization	Helpful dialogue	Model	Harmless dialogue
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RLHF vs SFT	73%	64%	RLHF	76%
RLAIIF vs RLHF	50%	52%	RLAIIF	88%
Same-size RLAIIF vs SFT	68%	—		
d-RLAIIF vs SFT	74%	66%		
d-RLAIIF vs Same-size RLAIIF	60%	—		

Results

D-RLAIF

- Annotators prefer d-RLAIF over same-size RLAIF 60% of the time.
- We hypothesize that this improvement is a result of directly querying the AI labeler for preferences rather than first distilling its preferences into a RM, as well as circumventing the “staleness” issue.
- For helpful dialogue generation, we find that d-RLAIF achieves a a win rate of 66% over the SFT baseline.
- Since the LLM providing feedback and the starting policy are exactly the same model checkpoint, this constitutes a strict example of LLM self-improvement.

Table 1: **Left side:** Win rates for pairs of policies on the summarization and the helpful dialogue tasks. **Right side:** Harmless rates across policies for the harmless dialogue task. All numbers are based on human evaluation.

Win Rate			Harmless Rate	
Comparison	Summa- rization	Helpful dialogue	Model	Harmless dialogue
RLAIF vs SFT	71%	63%	SFT	64%
RLHF vs SFT	73%	64%	RLHF	76%
RLAIF vs RLHF	50%	52%	RLAIF	88%
Same-size RLAIF vs SFT	68%	—		
d-RLAIF vs SFT	71%	66%		

Results

Prompting Techniques

- We experiment with three types of prompting variations **preamble specificity**, **chain-of-thought reasoning**, and **incontext learning**

Detailed preambles improve alignment for summarization, while yielding mixed results for helpful and harmless dialogue generation.

Chain-of-thought reasoning generally improves alignment. For summarization, the improvement is consistent.

Surprisingly, we observe that in-context learning only improves alignment for harmless dialogue generation

Table 2: We observe that eliciting chain-of-thought reasoning tends to improve AI labeler alignment, while few-shot prompting and detailed preambles have mixed effects across tasks. Above, “Help.” and “Harm.” refer to helpfulness and to harmlessness, respectively.

Prompt	AI Labeler Alignment		
	Summary	Help.	Harm.
Base 0-shot	76.1%	67.8%	69.4%
Base 1-shot	76.0%	67.1%	71.7%
Base 2-shot	75.7%	66.8%	72.1%
Base + CoT 0-shot	77.5%	69.1%	70.6%
Detailed 0-shot	77.4%	67.6%	70.1%
Detailed 1-shot	76.2%	67.6%	71.5%
Detailed 2-shot	76.3%	67.3%	71.6%
Detailed 8-shot	69.8%	—	—
Detailed + CoT 0-shot	78.0%	67.8%	70.1%
Detailed + CoT 1-shot	77.4%	67.4%	69.9%
Detailed + CoT 2-shot	76.8%	67.4%	69.2%

Results

Size of LLM Labeler

- A trend consistent with scaling behaviors observed in other work (Kaplan et al., 2020). One contributing factor to this trend, apart from a decrease in general model capability, may be that smaller LLMs display greater position bias
- From another angle, these results also suggest that scaling up AI labeler size may produce even higher quality preference labels.

Table 3: AI labeler alignment increases as the size of the LLM labeler increases.

Model Size	AI Labeler Alignment
PaLM 2 L	78.0%
PaLM 2 S	73.8%
PaLM 2 XS	62.7%

Qualitative Observations

we identified a few patterns where they sometimes diverged.

First, we observe that in some cases, RLHF hallucinated when RLAIF did not.

Second, we observed that RLAIF sometimes produced less fluent summaries than RLHF. For instance

In small-scale evaluation on 70 examples, where human annotators were asked to blindly rank RLHF and RLAIF summaries in terms of accuracy, coverage, and coherence, the difference in scores was not statistically significant.

Conclusion

- In head-to-head comparisons, RLAIIF and RLHF are preferred at similar rates by humans
- we demonstrate evidence of LLM self-improvement by showing that RLAIIF is effective even when the LLM labeler is the same size as the policy, or even the exact same checkpoint as the initial policy.
- we also direct-RLAIIF, which directly prompts the LLM labeler to provide rewards during RL, outperforming the canonical RLAIIF setup that first distills LLM preferences into a separate RM.
- we study the impact of various AI labeling techniques on alignment to human preferences.